

Riad Dajani

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SUMMARY

ML Researcher and founder with 4+ years building toward one long-term goal: enhancing cognitive bandwidth and extending the reach of human consciousness. At Johns Hopkins, I study brain-aligned self-supervised models — understanding how intelligence emerges from prediction. At Superpower Social, which I co-founded, I shipped production AI systems that adapt in real time to how people actually behave. Research that becomes tools. Tools that learn from people. All of it pointing in the same direction.

SKILLS

- **Deep Learning & AI Research:** PyTorch (DDP, Lightning), Self-Supervised Learning (JEPA, DINO, VICReg), Neural Decoding (fMRI, ECoG), Vision Transformers (ViT, I3D, VideoMAE), Multimodal Models (CLIP, LLaVA), BiLSTM-Attention, GNN, Contrastive & Generative Objectives, Computer Vision, Video Understanding, VLAs, World Modeling, RL, LLMs, Brain-Model Alignment (RSA, linear probing), Representational Geometry, Model Interpretability & Mechanistic Analysis, BCI / Neural Interface Systems, Human-Aligned Video & Language Models
- **ML Engineering & Applied AI:** ML Pipelines (training, eval, deployment), Feature Engineering, Rapid Prototyping, Production Model Integration, TensorFlow, SLURM, DDP, Decision-First Architecture, Experiment Design & Tracking (W&B), LLM Routing & Cost Optimization, Online Learning (SGD), Real-Time Inference Pipelines, Self-Improving Prompt Systems, RAG Pipelines, Fine-Tuning, Agentic Workflow Evaluation, Human-in-the-Loop System Design, Feedback Loop Engineering, Feature Engineering, Rapid Prototyping, Hugging Face, timm
- **Data & Infrastructure:** S3-Optimized Video & Feature Stores, Scalable Data Ingestion & Labeling, Large-Scale Distributed Processing, Snowflake / Postgres, Online SGD, Dataset Curation & Versioning, Real-Time Webhook Ingestion, Microservices Architecture, REST API Integration (Meta, Shopify), Analytics Dashboards
- **Systems:** Python, C++, TypeScript, JavaScript, AWS (EKS, S3, Batch), GCP, Slurm HPC, CUDA, Cluster Computing, Full-Stack Development
- **Startup & Business:** 0 to 1 Product Development, First Principles, Rapid Iteration Cycle, User Research & Discovery, Founder-led Sales, Growth Marketing, Unit Economics & Pricing Strategy, Contract Negotiation, Supply Chain Management, Cross-functional Team Leadership, Hiring & Technical Assessment Design, Resource-Constrained Decision-Making

EXPERIENCE

Handshake AI Fellowship

Aug 2025 - Feb 2026

AI Trainer & Reviewer – ML, CS & Math Expert

Remote

- Produced 400-level undergraduate test questions based in CS and Mathematics to uncover failure modes VLMs.
- Corrected Agentic workflows including the production of ML code when displaying rogue behavior.
- Reviewed over 100 other fellow-made corrections.

Johns Hopkins University, Isik Computational Cognitive Neuroscience Lab

2024 - Mar 2026

Graduate Researcher

Baltimore, MD, USA

Interdisciplinary focus on computational cognitive neuroscience, deep learning, and brain-computer interfaces, integrating theory and large-scale modeling to study neural representations of language and perception.

- Created ***BrainLang***, a neural decoding system translating **fMRI activity into natural language** during speech comprehension, achieving **~47% decoding accuracy above chance**; extended in v2 with improved preprocessing and training stability for robust cross-subject generalization
- Modeled visual perception as **probabilistic (Bayesian) inference**, applying **Markov Random Fields, Structured Latent SVMs, and Hierarchical Compositional Models to object detection, segmentation, and recognition tasks**.
- Engineered scalable ML training infrastructure for large multimodal models, including **distributed data loading**, mixed-precision training, and **SLURM-based job orchestration**, enabling efficient experimentation across high-dimensional neuroscience and behavioral datasets.
- Implemented an **I3D-based** pipeline to extract and quantize spatiotemporal embeddings from the **ASL-Citizen corpus for phonological analysis**, contributing to the standardization of deep learning evaluation methods in sign language modeling, enhancing model accuracy and consistency.
- Contributed to the development of a Dynamic Social Interaction Model by creating a joint predictive architecture for future latent states, **improving downstream performance by ~15% while preventing embedding collapse**.
- Derived a formal axiomatic framework for Kolmogorov-Arnold Networks (KANs), proving that learnable univariate edge functions can replace traditional weight matrices to yield more interpretable and parameter-efficient neural architectures

Superpower Social

Nov 2023 - Oct 2025

Founder

Toronto, ON, Canada

Founded, led and scaled an AI startup building real-time social media moderation and response systems for brands and creators at scale.

- Founded and scaled an AI startup delivering real-time social media moderation and response systems, serving **300+ brands** and processing **millions of interactions** across major platforms.
- Built and deployed **online-learning ML systems (SGD + embeddings)** for behavior prediction and moderation, achieving **sub-10 ms inference** and significantly reducing API usage and cost.
- Leveraged UI telemetry as human feedback to iteratively refine prompt and user-behavior prediction models, **enhancing suggestion acceptance to 87% and auto-engage accuracy to 78%**.
- Managed enterprise-level data processing with over 1 TB of data per month **at 99.95% uptime**, integrating distributed systems and large-scale data processing techniques in production ML scenarios.
- Led end-to-end system design and development and a **4-person engineering team**, managing large-scale data pipelines (- **1TB+/month, 99.95% uptime**) across full-stack, ML infrastructure, and cloud deployment

Nasdaq DataLink

May 2022 - Sep 2022

Data Analytics Engineer Intern

Toronto, ON, Canada

Designed and optimized backend systems for massive data-intensive applications, contributing to the development of scalable data pipelines and integrating Python-based tools to enhance automation processes.

- Collaborated with cross-functional teams to deliver data-driven risk assessments and performance dashboards that support experimental prototype evaluation in fast-paced environments.
- Developed demo predictive ML models integrated into analytics platforms, enhancing search functionality for the DataLink data market platform and exemplifying end-to-end ownership of tool development.

Khoulas Clothing

Sep 2018 - Sep 2024

Founder

Managed all aspects of a benefit clothing business aimed at sustainably raising money for children in need, including product design, supplier sourcing, manufacturing, procurement, logistics, warehousing, shipping, retail partnerships, sales, marketing, and financial accounting.

- Negotiated contracts to reduce unit costs by 18% and implemented demand forecasting to decrease stockouts by 15%.
- Scaled customer base to 7,000 through targeted marketing and dynamic pricing strategies.

EDUCATION

Johns Hopkins University

Jan 2024 - May 2025

Master of Arts, Computational Cognitive Science

Baltimore, MD, USA

- **GPA:** 4.0 GPA
- **Coursework:** Foundations of Neural Network Theory, Vision as Bayesian Inference, Deep Learning for Cognitive Neuroscience, Computational Cognitive Neuroscience of Vision, Theory, Modeling & Coding Neural Activity, Research Seminar in Cognitive Neuroscience & Machine Learning, Responsible Conduct of Research, Live subject EEG experimentation

University of Waterloo

Sep 2019 - Mar 2023

Bachelor of Arts, Global Business and Digital Arts

Waterloo, ON, Canada

- **GPA:** 87% GPA
- **Achievements:** Dean's Honors List
- **Coursework:** Perception, Quantitative Methods, Computational Neuroscience Methods, Linguistics, Physiological Psychology, User Experience Research, Cognitive Science

PROJECTS

DeepJuice (Model Injection) | <https://github.com/ColinConwell/DeepJuiceDev/tree/injection>

Dec 2025

- Contributed to **DeepJuice (AI Research toolkit)** with a **model-agnostic inference-time injection system**, enabling reversible perturbations of internal activations to induce controlled high-entropy, low-precision representational states.
- Implemented **activation, connectivity, and attention-level injections** (entropy modulation, precision relaxation, stochastic mixing, excitability gain, receptive-field expansion) to disrupt rigid priors without retraining.
- Designed a **unified injection API** for consistent benchmarking across language, image, and video models, supporting transient “altered states” without permanent parameter changes.

Community Notes + SIGreg | <https://github.com/riaddajani/communitynotes/tree/SIGreg>

- Enhanced Community Notes ranking by modifying matrix factorization with **SIGreg**, preserving multi-factor diversity and preventing collapse (variance ratio ~0.7)
- Discovered stable, interpretable latent factors (e.g., political polarity, language, tone) with strong **temporal and cross-seed consistency** ($r = 0.55-0.66$)
- Improved robustness to manipulation by **reducing brigade attacks 24–43%** and enabled downstream signal extraction (e.g., humor detection, AUC = 0.758)

BrainLang | <https://archive.org/details/brainlang-report>

Johns Hopkins University

- Built **BrainLang**, a simple seq2seq BiLSTM + attention model to decode continuous language from fMRI signals, achieving **47% word-level accuracy** on natural narrative data

- Designed an end-to-end pipeline combining **UMAP dimensionality reduction + temporal modeling**, improving tractability and capturing multi-timescale neural dynamics from high-dimensional brain data
- Demonstrated feasibility of **non-invasive neural language decoding**, producing coherent phrase-level reconstructions and advancing high-bandwidth brain-computer communication systems

Audio in, Digit out Model | <https://github.com/riaddajani/FSDD-Digits>

Aug 2025

- **Built a fast, low-latency spoken-digit recognizer** using a compact MFCC-based CNN, achieving **99%+ accuracy and macro-F1 on the FSDD benchmark** with millisecond-scale CPU inference.
- **Implemented early-exit streaming inference and few-shot live enrollment**, enabling confident predictions before utterance completion and rapid adaptation to new speakers, languages or speech domains.
- **Added real-time interpretability and diagnostics**, including per-epoch confusion matrices and live saliency over MFCC bands to explain model decisions during deployment.

Eyes Closed/Open EEG

<https://colab.research.google.com/drive/1NNIO9PMZU-9Yol00UQrguqnbPBMzgKFe?usp=sharing>

University of Waterloo

- Built an end-to-end **EEG-based brain-computer interface (BCI)** pipeline, classifying eyes-open vs. eyes-closed states from occipital signals and translating predictions into real-time cursor control.
- Engineered signal processing and feature extraction pipeline (**FFT + power spectral density**) to isolate alpha-band activity (8–13 Hz) as a reliable neural marker of visual state.
- Implemented a **from-scratch neural network (NumPy)** with manual backpropagation, achieving clear class separation and demonstrating a full brain → model → action loop for neural interfacing.

Self Improving Prompt | <https://github.com/riaddajani/self-improving-prompt>

Mar 2025

Superpower Social

- **Built a self-improving prompt optimization loop for LLMs** that generates persona-driven social reply prompts, then iteratively tests, critiques, patches (unified diffs), then re-tests to reduce error against ground-truth responses (measured via embedding cosine distance).
- **Shipped a production-ready Python package + CLI** (refactored from a notebook) with configurable hyperparameters (early stopping/patience, exemplar sampling, thresholds) and validation-loss tracking/plotting; prototype later informed Superpower Social's enterprise persona-alignment system.

Robotic Painting | https://www.youtube.com/watch?v=hMDyhlG_PVI

University of Waterloo

- A portrait I made of the department dean with animatronic eyes. Using a Raspberry Pi and OpenCV, it detects and tracks the faces of unsuspecting students passing by.